Path 2: Cyber Security 101 > Module 3: Windows and AD Fundamentals > Room 2: Windows Fundamentals 2

https://tryhackme.com/room/windowsfundamentals2x0x

Windows Fundamentals 2:
In part 2 of the Windows Fundamentals module, discover more about System Configuration, UAC Settings, Resource Monitoring, the Windows Registry and more..

>> Task 1: Introduction

We will continue our journey exploring the Windows operating system.

In **Windows Fundamentals 1**, we covered the desktop, the file system, user account control, the control panel, settings, and the task manager.

This module will attempt to provide an overview of some other utilities available within the Windows operating system and different methods to access these utilities.

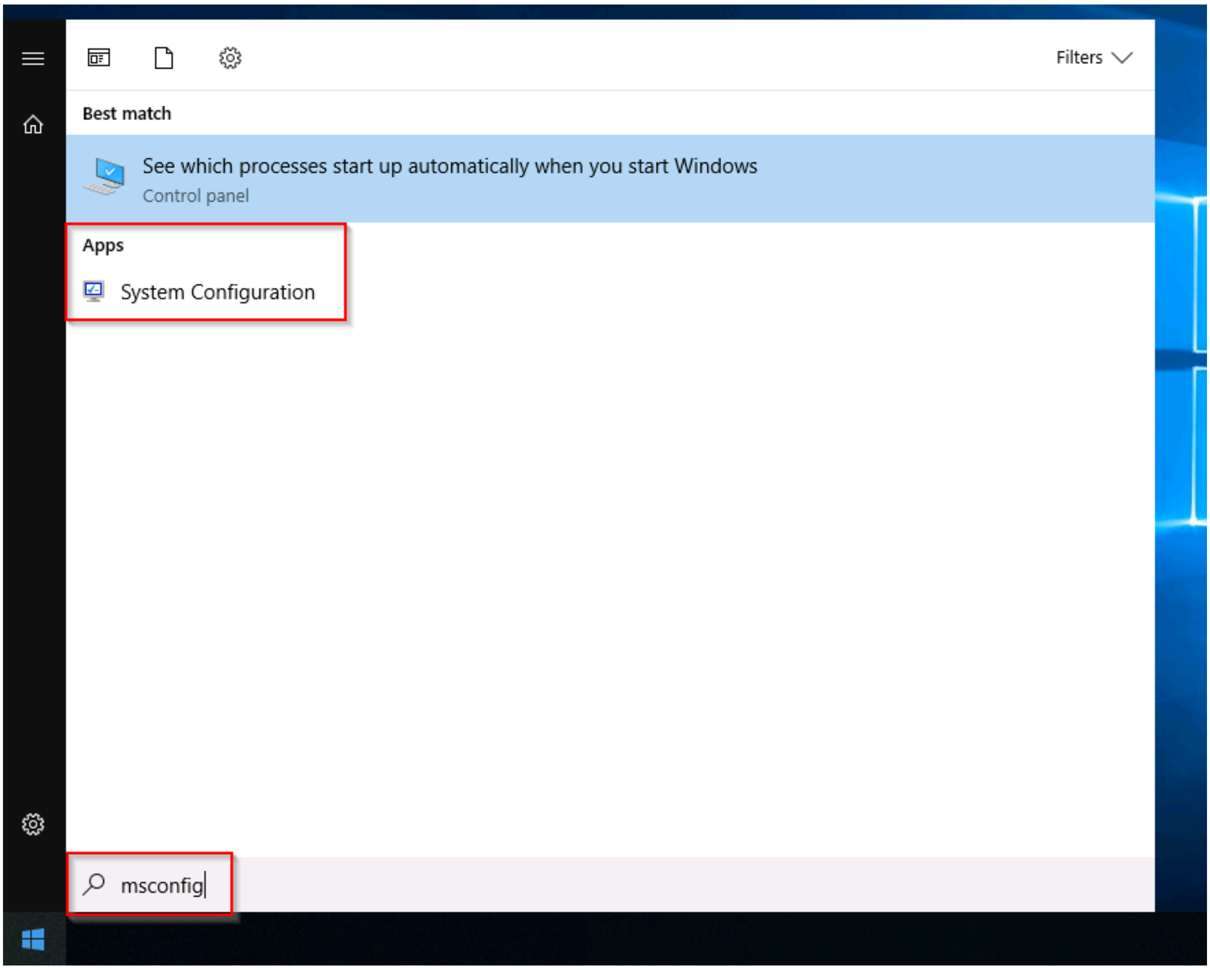
>> Task 2: System Configuration and Advanced System Settings

- System Configuration:

The **System Configuration** utility (**MSConfig**) is for advanced troubleshooting, and its main purpose is to help diagnose startup issues.

Reference the following document here (https://learn.microsoft.com/en-us/troubleshoot/windows-client/performance/system-configuration-utility-troubleshoot-configuration-errors) for more information on the System Configuration utility.

There are several methods to launch **System Configuration**. One method is from the Start Menu.



Note: You need local administrator rights to open this utility.

The utility has five tabs across the top. Below are the names for each tab. We will briefly cover each tab in this task.
- General
- Boot
- Services
- Startup
- Tools

In the **General** tab, we can select what devices and services for Windows to load upon boot. The options are: **Normal**, **Diagnostic**, or **Selective**.

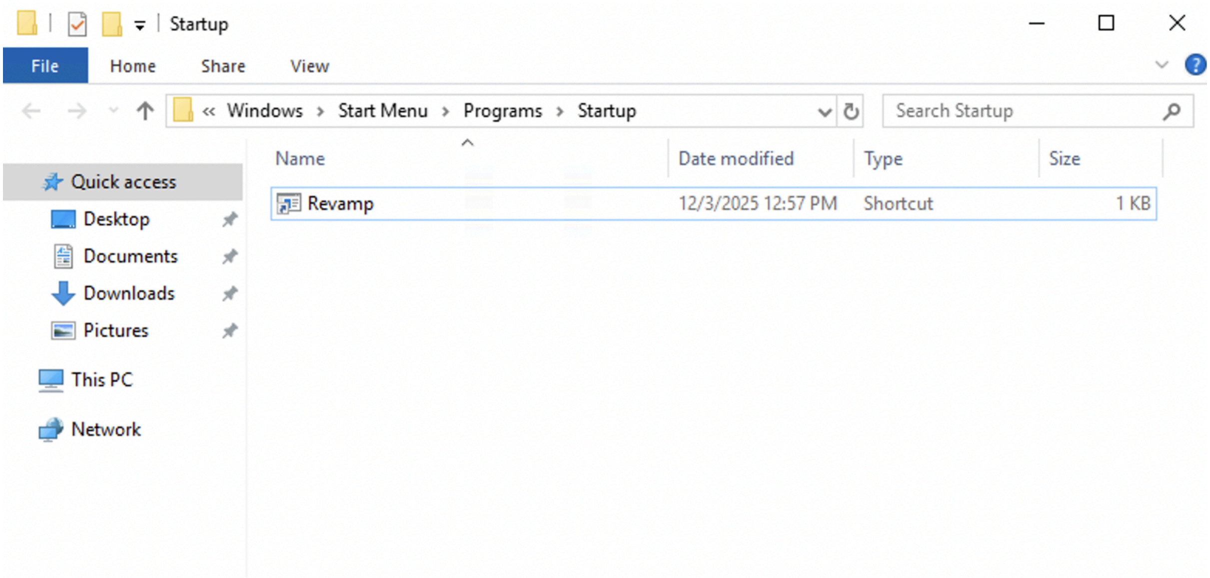
In the **Boot** tab, we can define various boot options for the Operating System.

The **Services** tab lists all services configured for the system regardless of their state (running or stopped). A service is a special type of application that runs in the background.

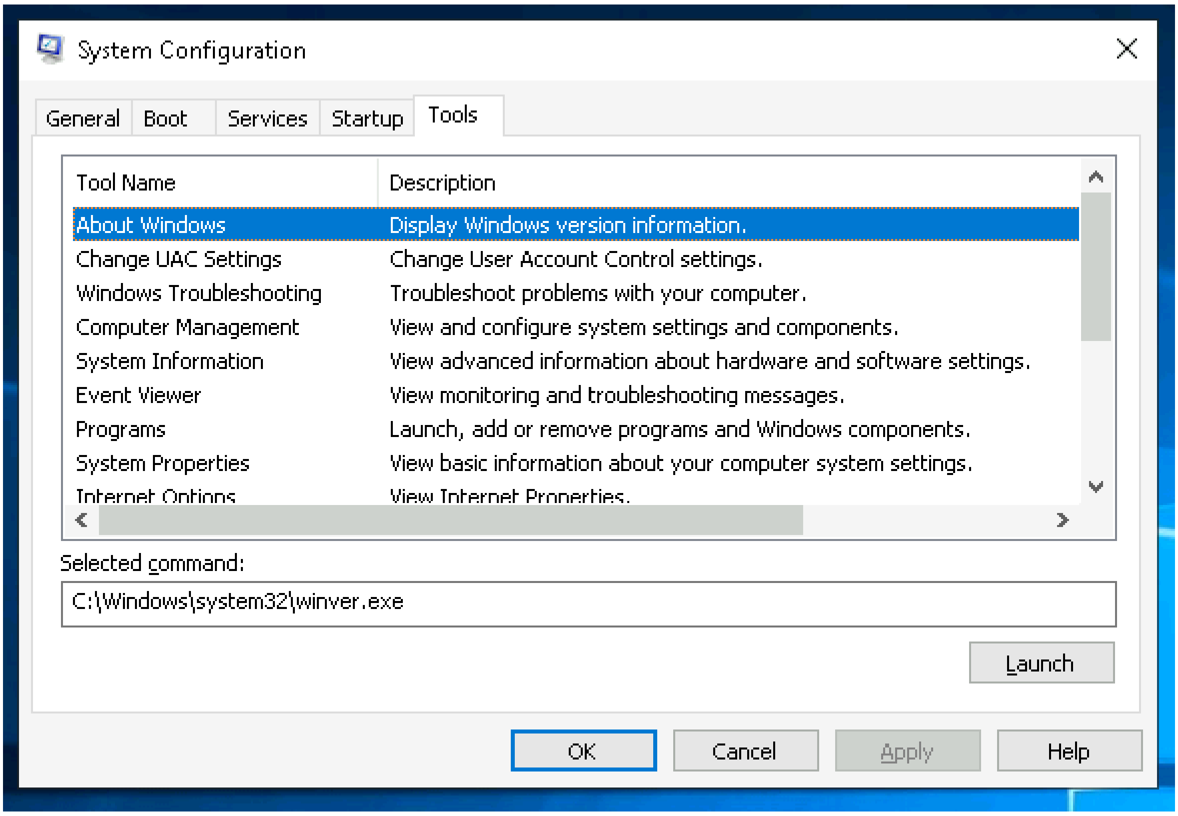
In the **Startup** tab, you won't see anything interesting. To manage startup items, use the Startup section of Task Manager.

As you can see, Microsoft advises using **Task Manager** (taskmgr) to manage (enable/disable) startup items. The System Configuration utility is NOT a startup management program.

If you open the Task Manager for the attached VM, you will notice that it doesn't display a Startup tab. You will also not see anything in the Startup tab inside the **msconfig** utility, as shown above. This is because the attached machine is a Windows server, and Windows servers handle startup applications differently than Windows client systems. Unlike Windows 10 or 11, you will not see startup programs in **Task Manager** or in the Startup tab of **msconfig**. On these Windows server machines, the only reliable way to view user-level startup items is through the Startup folder itself. You can access it by pressing **Win + R**, which opens the Run Dialog, typing **shell:startup**, and then pressing Enter. This will display all startup programs as shortcuts or executables that are configured to run automatically the next time a user logs in. This is where you can verify applications that are configured to launch at startup.



Now coming back to the **msconfig**. There is a list of various utilities (tools) in the Tools tab here that we can run to configure the operating system further. There is a brief description of each tool to provide some insight into what the tool is for.

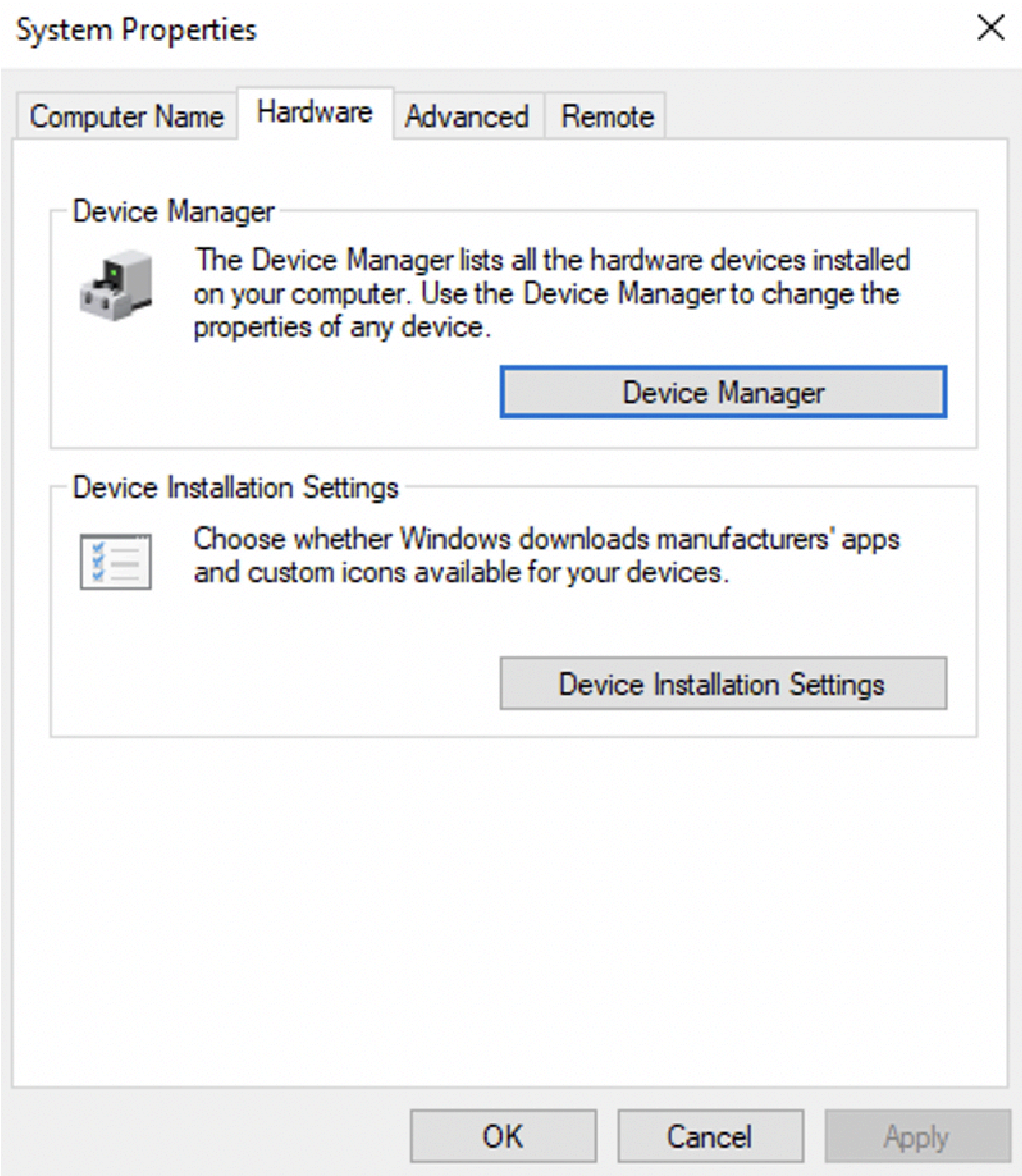


Notice the **Selected command** section. The information in this textbox will change per tool.

To run a tool, we can use the command to launch the tool via the run prompt, command prompt, or by clicking the **Launch** button.

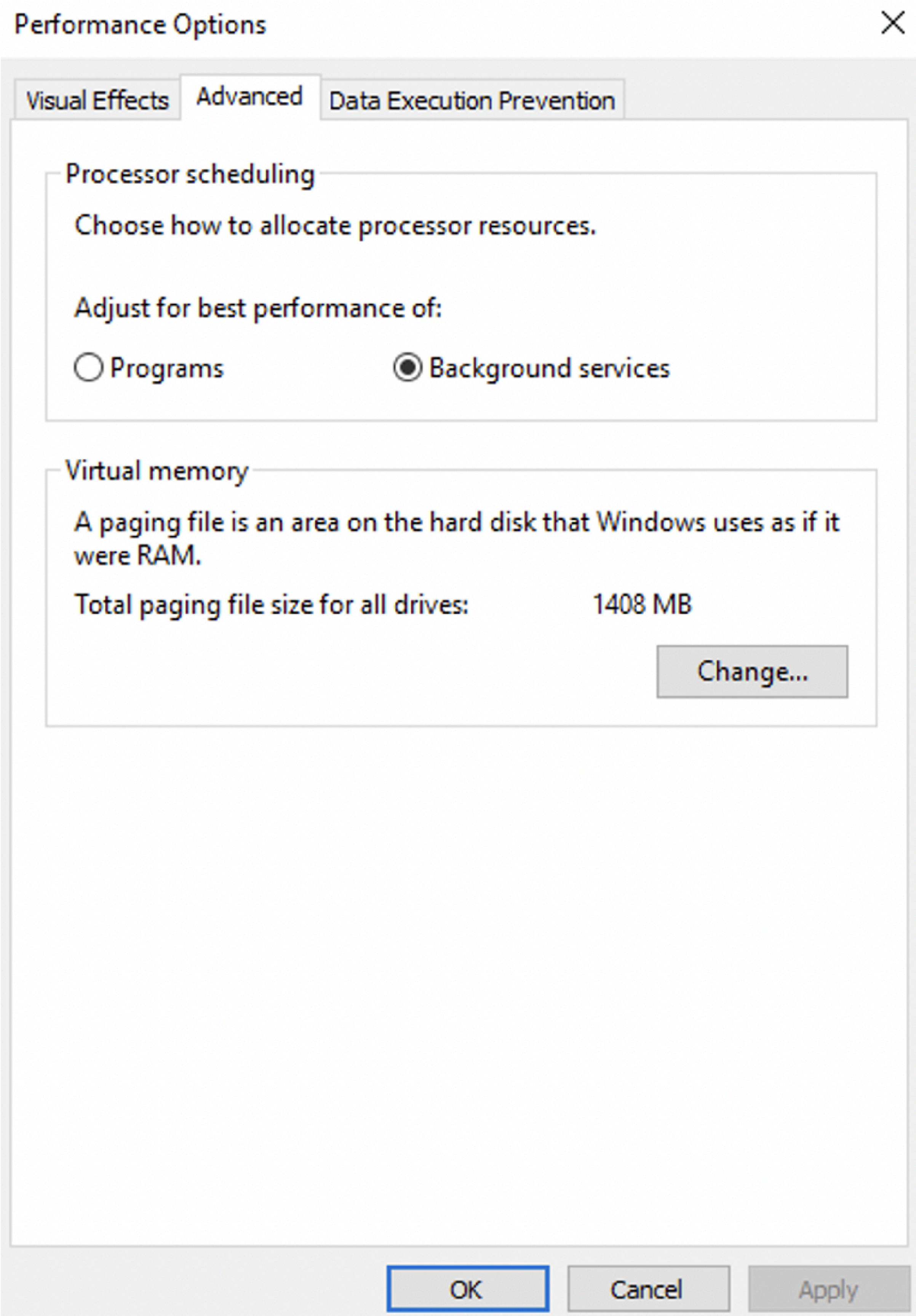
- Advanced System Settings:

Windows gives you some additional configuration settings as well, which you can use to control the performance behavior and system recovery. To access this option, you can search for **View advanced system settings** in your search bar and open it. This will open a **System Properties** panel as shown below:



Windows uses a page file as an extra virtual memory space when the physical RAM becomes full. This helps to prevent slowdowns or application crashes when the system runs out of memory. You can view or modify the page file by navigating to the **Advanced** option at the top and clicking **Settings** under the **Performance** tab.

So after clicking the Settings, you will get the Performance Options window, as can be seen below:


Performance Options
Visual Effects
Advanced
Data Execution Prevention
Processor scheduling
Choose how to allocate processor resources.
Adjust for best performance of:
Programs
Background services
Virtual memory
A paging file is an area on the hard disk that Windows uses as if it were RAM.
Total paging file size for all drives:
1408 MB
Change...
OK
Cancel
Apply

In this Performance tab, the **Advanced** option can also tell you about the page file size configured for the drives. In this case, it's **1048 MB**. The other settings here can give you the following information:

- The drive where the page file is stored
- The initial size (MB)
- The maximum size
- Whether Windows manages the size automatically

There is another cool configuration that you can find in the Advanced System Settings. It is known as Startup and Recovery. Windows can create a crash dump file whenever it encounters a critical error, such as a Blue Screen of Death. This crash dump helps the administrators or analysts understand what went wrong during the crash. You can view or modify the crash dump settings by navigating to the **Advanced** option at the top and then clicking **Settings** under the **Startup and Recovery** section.

So after clicking the settings, you will see the **Startup and Recovery** window.

Here, you will find different settings for the startup and recovery. The **Write debugging information** dropdown tells you the type of crash dump configured for the system. Windows supports different dump types, such as:

- Automatic memory dump
- Kernel memory dump
- Small memory dump (256 KB)
- Complete memory dump
- None

This setting shows how much information Windows will save in the crash dump when a system crash occurs.

Win+R: msconfig

Q1) What is the name of the service that lists Systems Internals as the manufacturer?
Answer: PsShutdown

Q2) Whom is the Windows license registered to?
Answer: Windows User

Q3) What is the command for Windows Troubleshooting?
Answer: C:\Windows\System32\control.exe /name Microsoft.Troubleshooting

Q4) What command will open the Control Panel? (The answer is the name of .exe, not the full path)
Answer: control.exe

>> Task 3: Change UAC Settings

We're continuing with Tools that are available through the System Configuration panel.

User Account Control (UAC) was covered in great detail in **Windows Fundamentals 1**.

The UAC settings can be changed or even turned off entirely (not recommended). You can move the slider to see how the setting will change the UAC settings and Microsoft's stance on the setting.

This slider has four security levels, each of which controls how Windows alerts you when apps or users try to make changes at the system level. They fall into four standard categories as explained below:

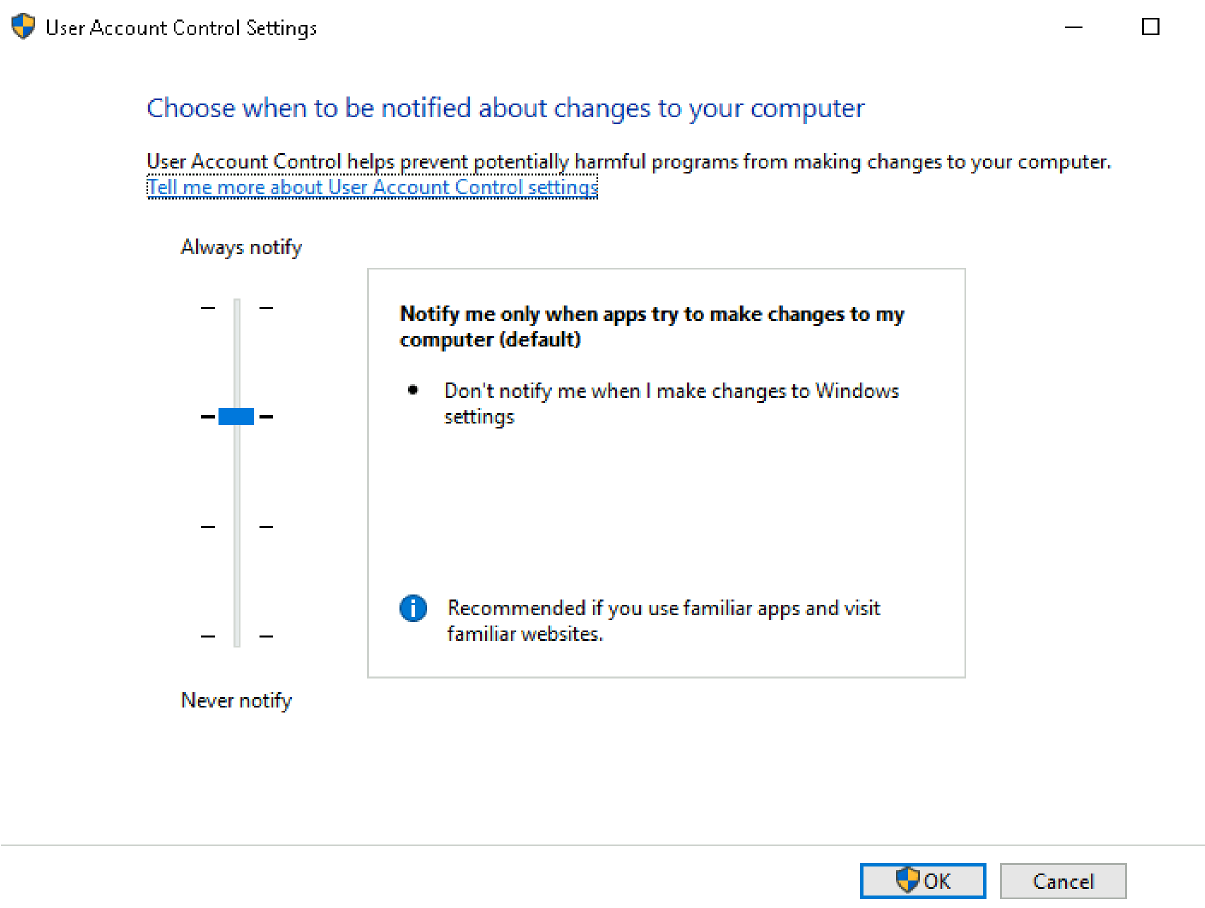
- **Always notify**: This is the highest security. Windows notifies you whenever any apps or you yourself try to make changes, and the desktop dims (Secure Desktop).

- **Notify for apps**: Windows notifies only when apps try to make changes, but not when you change Windows settings. This option is enabled by default.

- **Notify without dimming**: Same as above (Notify for apps), but this time the screen does not dim.

- **Never notify**: Notifications are turned off. Windows won’t warn you about any changes made by you or any apps.

You can find the current level by looking at the position of the slider in the **User Account Control Settings** window.



Q1) What is the command to open User Account Control Settings? (The answer is the name of the .exe file, not the full path)
Answer: UserAccountControlSettings.exe

>> Task 4: Computer Management

We're continuing with tools that are available through the **System Configuration** panel.

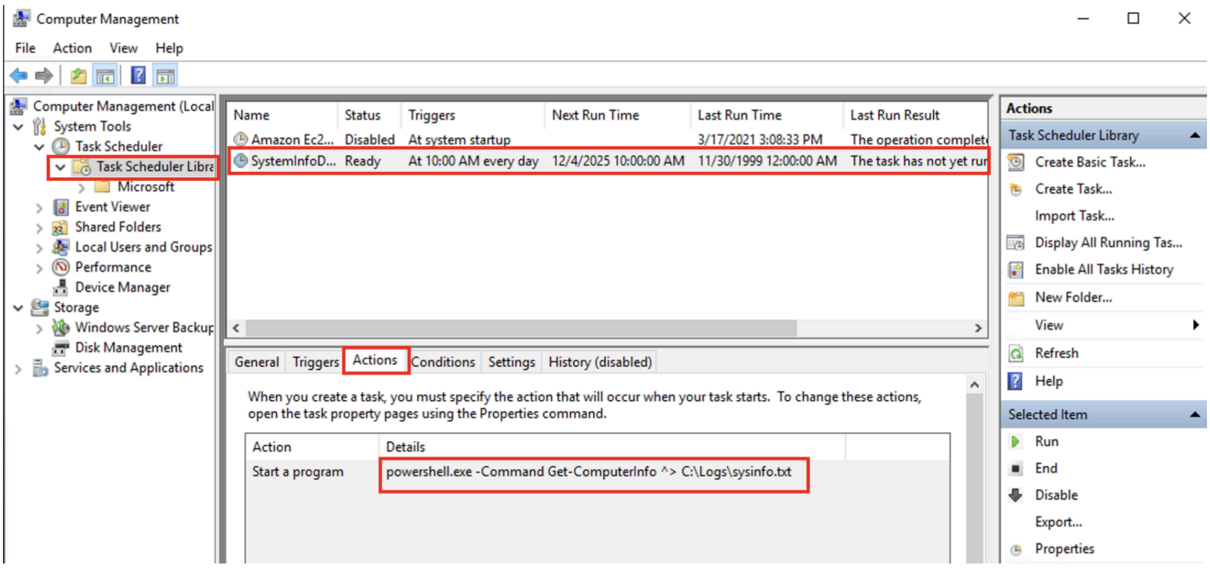
The **Computer Management** (compmgmt) utility has three primary sections: **System Tools, Storage,** and **Services and Applications**.

- System Tools:

Let's start with **Task Scheduler**. Per Microsoft, with Task Scheduler, we can create and manage common tasks that our computer will carry out automatically at the times we specify.

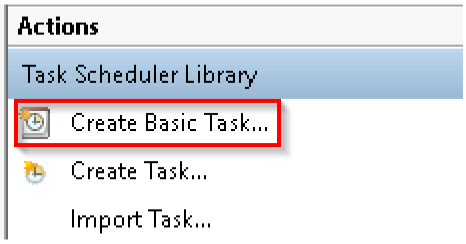
A task can run an application, a script, etc., and tasks can be configured to run at any point. A task can run at log in or at log off. Tasks can also be configured to run on a specific schedule, for example, every five mins.

To view the scheduled tasks that are present on the system, click **Task Scheduler Library**. This will display all the scheduled tasks of the system. You can click on any of them to view their details. The screenshot below shows a scheduled task named **SystemInfoDailyLog** configured to run **every day at 10:00 AM**. Here, you will see the program or command that will run when the task is triggered.



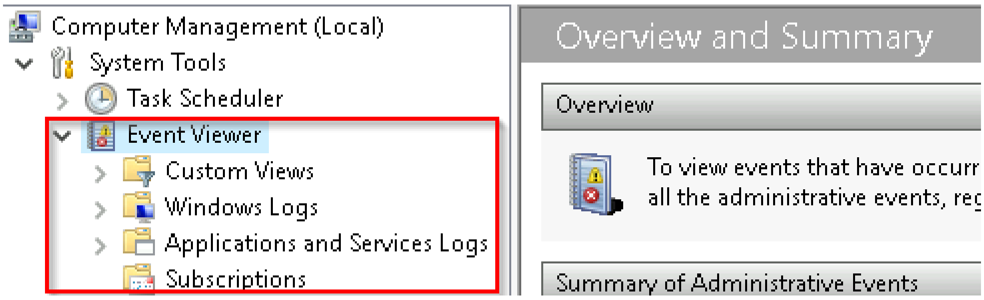
It is also important to note that some scheduled tasks are not recurring and are made to run just once at a specific time. In this case, we would see something like **At 2:50 PM on 6/15/2025** as the trigger.

To create a basic task, click on **Create Basic Task** under **Actions** (right pane).



Next is **Event Viewer**.

Event Viewer allows us to view events that have occurred on the computer. These records of events can be seen as an audit trail that can be used to understand the activity of the computer system. This information is often used to diagnose problems and investigate actions executed on the system.



Event Viewer has three panes:
- The pane on the left provides a hierarchical tree listing of the event log providers. (as shown in the image above)
- The pane in the middle will display a general overview and summary of the events specific to a selected provider.
- The pane on the right is the actions pane.

There are five types of events that can be logged. Below is a table from docs.microsoft.com (https://learn.microsoft.com/en-us/windows/win32/eventlog/event-types) providing a brief description for each.

The following table describes the five event types used in event logging.

Event type	Description
Error	An event that indicates a significant problem such as loss of data or loss of functionality. For example, if a service fails to load during startup, an Error event is logged.
Warning	An event that is not necessarily significant, but may indicate a possible future problem. For example, when disk space is low, a Warning event is logged. If an application can recover from an event without loss of functionality or data, it can generally classify the event as a Warning event.
Information	An event that describes the successful operation of an application, driver, or service. For example, when a network driver loads successfully, it may be appropriate to log an Information event. Note that it is generally inappropriate for a desktop application to log an event each time it starts.
Success Audit	An event that records an audited security access attempt that is successful. For example, a user's successful attempt to log on to the system is logged as a Success Audit event.
Failure Audit	An event that records an audited security access attempt that fails. For example, if a user tries to access a network drive and fails, the attempt is logged as a Failure Audit event.

Event Type: Error
Description: An event that indicates a significant problem such as loss of data or loss of functionality. For example, if a service fails to load during startup, an Error event is logged.

Event Type: Warming
Description: An event that is not necessarily significant, but may indicate a possible future problem. For example, when disk space is low, a Warning event is logged. If an application can recover from an event without loss of functionality or data, it can generally classify the event as a Warning event.

Event Type: Information
Description: An event that describes the successful operation of an application, driver, or service. For example, when a network driver loads successfully, it may be appropriate to log an Information event. Note that it is generally inappropriate for a desktop application to log an event each time it starts.

Event Type: Success Audit
Description: An event that records an audited security access attempt that is successful. For example, a user's successful attempt to log on to the system is logged as a Success Audit event.

Event Type: Failure Audit
Description: An event that records an audited security access attempt that fails. For example, if a user tries to access a network drive and fails, the attempt is logged as a Failure Audit event.

The standard logs are visible under **Windows Logs**. Below is a table from docs.microsoft.com (https://learn.microsoft.com/en-us/windows/win32/eventlog/eventlog-key) providing a brief description for each.

The event log contains the following standard logs as well as custom logs:

Log	Description
Application	Contains events logged by applications. For example, a database application might record a file error. The application developer decides which events to record.
Security	Contains events such as valid and invalid logon attempts, as well as events related to resource use such as creating, opening, or deleting files or other objects. An administrator can start auditing to record events in the security log.
System	Contains events logged by system components, such as the failure of a driver or other system component to load during startup.
CustomLog	Contains events logged by applications that create a custom log. Using a custom log enables an application to control the size of the log or attach ACLs for security purposes without affecting other applications.

Log: Application
Description: Contains events logged by applications. For example, a database application might record a file error. The application developer decides which events to record.

Log: Security
Description: Contains events such as valid and invalid logon attempts, as well as events related to resource use such as creating, opening, or deleting files or other objects. An administrator can start auditing to record events in the security log.

Log: System
Description: Contains events logged by system components, such as the failure of a driver or other system component to load during startup.

Log: CustomLog
Description: Contains events logged by applications that create a custom log. Using a custom log enables an application to control the size of the log or attach ACLs for security purposes without affecting other applications.

For more information about Event Viewer and Event Logs, please refer to the **Windows Event Log** room (https://tryhackme.com/room/windowseventlogs).

Shared Folders is where you will see a complete list of shares and folders shared that others can connect to.

Share Name	Folder Path	Type	# Client Connections	Description
ADMIN$	C:\Windows	Windows	0	Remote Admin
C$	C:\	Windows	0	Default share
IPC$		Windows	0	Remote IPC

In the above image, under Shares, are the default share of Windows, C$, and default remote administration shares created by Windows, such as ADMIN$.

As with any object in Windows, you can right-click on a folder to view its properties, such as Permissions (who can access the shared resource).

Under **Sessions**, you will see a list of users who are currently connected to the shares. In this VM, you won't see anybody connected to the shares.

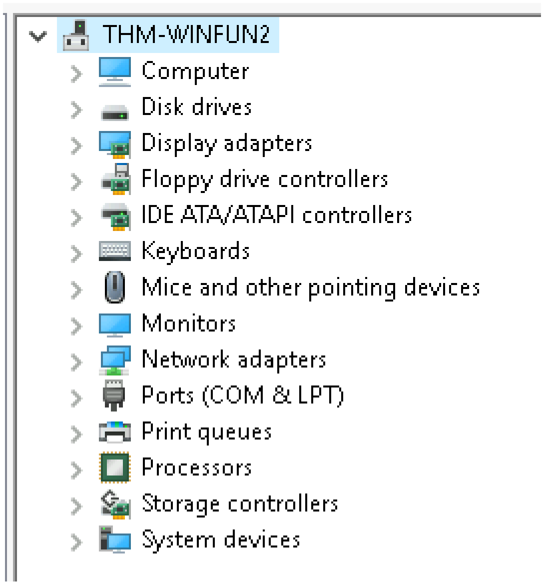
All the folders and/or files that the connected users access will list under **Open Files**.

The **Local Users and Groups** section you should be familiar with from Windows Fundamentals 1 because it's lusrmgr.msc.

In **Performance**, you'll see a utility called **Performance Monitor** (perfmon).

Perfmon is used to view performance data either in real-time or from a log file. This utility is useful for troubleshooting performance issues on a computer system, whether local or remote.

Device Manager allows us to view and configure the hardware, such as disabling any hardware attached to the computer.



Storage

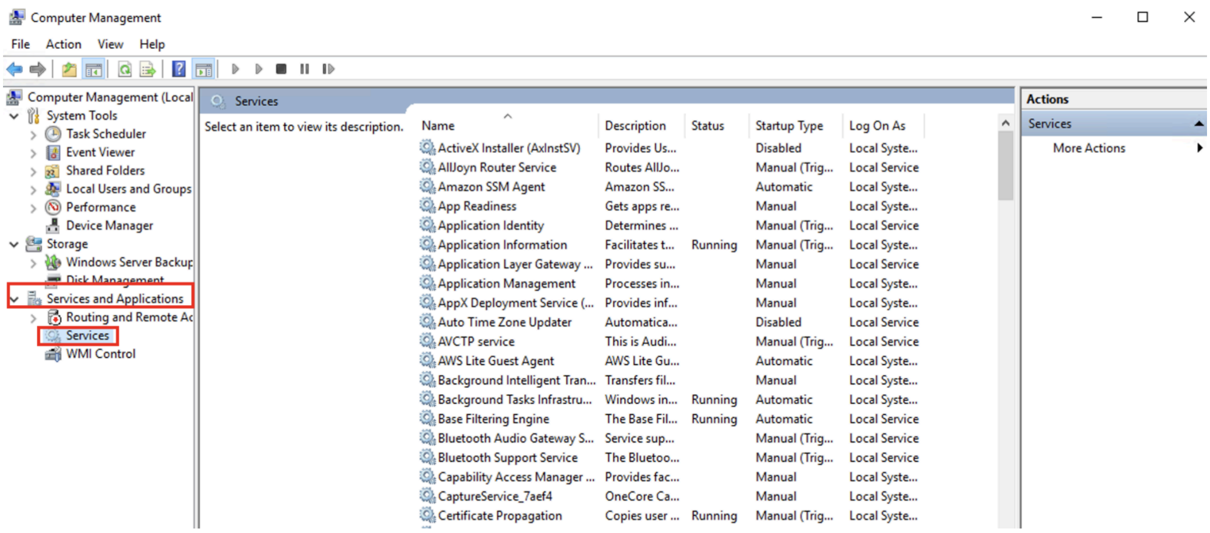
Under Storage is **Windows Server Backup** and **Disk Management**. We'll only look at Disk Management in this room.

Note: Since the virtual machine is a Windows Server operating system, there are utilities available that you will typically not see in Windows 10.

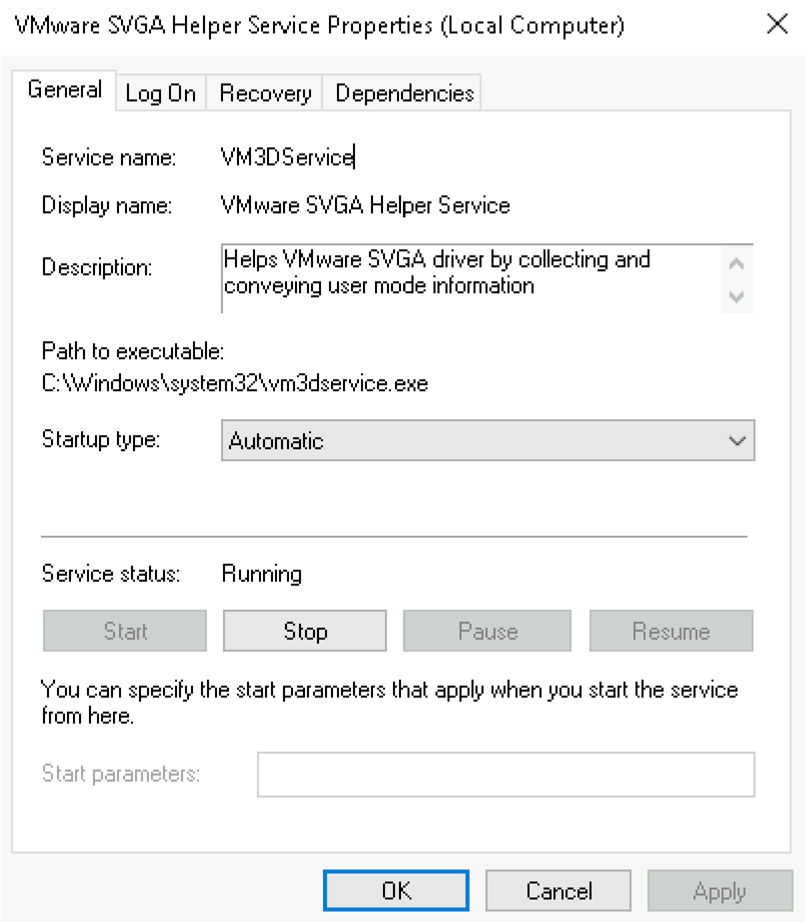
Disk Management is a system utility in Windows that enables you to perform advanced storage tasks. Some tasks are:
- Set up a new drive
- Extend a partition
- Shrink a partition
- Assign or change a drive letter (ex. E:)

Services and Applications

Recall from the previous task, a service is a special type of application that runs in the background. You can see all the services and their statuses by clicking the Services button given under the Services and Applications section.



The services shown above have their display names, status, and other values. If you want to get more information about any service, right-click on the service and click **properties**. Here, you will see additional details, such as the service name (which differs from the display name), the path to its executable, its startup type, and other relevant information.



There is a field known as Startup type in a service's Properties window, as shown above. It determines how and when the service is configured to start. We can set a service to **Automatic**, which means it starts every time the system boots, or **Manual**, which means it only starts when another process or user triggers this service, or **Disabled**, which means it should not run at all.

WMI Control configures and controls the **Windows Management Instrumentation** (WMI) service.

Per Wikipedia, "WMI allows scripting languages (such as VBScript or Windows PowerShell) to manage Microsoft Windows personal computers and servers, both locally and remotely. Microsoft also provides a command-line interface to WMI called Windows Management Instrumentation Command-line (WMIC)."

Note: The WMIC tool is deprecated in Windows 10, version 21H1. Windows PowerShell supersedes this tool for WMI.

Q1) What is the command to open Computer Management?
(The answer is the name of the .msc file, not the full path)
Answer: compmgmt.msc

Q2) When is the **npcapwatchdog** scheduled task set to run at?
Answer: At system startup

Q3) What is the name of the hidden folder that is shared?
Answer: sh4r3dF0Ld3r

>> Task 5: System Information

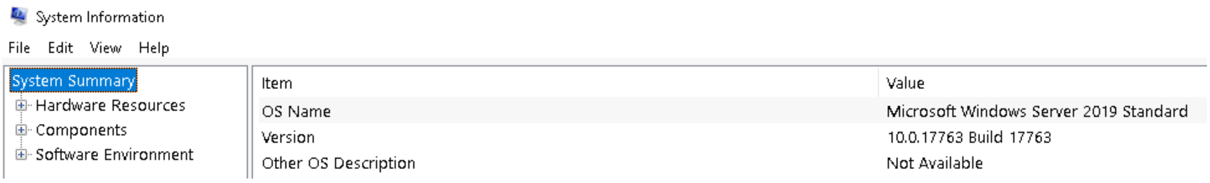
We're continuing with Tools that are available through the System Configuration panel.

What is the **System Information** (msinfo32) tool?

Per Microsoft, "Windows includes a tool called Microsoft System Information (Msinfo32.exe). This tool gathers information about your computer and displays a comprehensive view of your hardware, system components, and software environment, which you can use to diagnose computer issues."

The information in **System Summary** is divided into three sections:
- **Hardware Resources**
- **Components**
- **Software Environment**

System Summary will display general technical specifications for the computer, such as processor brand and model.



The information displayed in **Hardware Resources** is not for the average computer user. If you want to learn more about this section, refer to the official Microsoft page (https://learn.microsoft.com/en-us/windows-hardware/drivers/kernel/hardware-resources).

Under **Components**, you can see specific information about the hardware devices installed on the computer. Some sections don't show any information, but some sections do, such as **Display** and **Input**.

In the **Software Environment** section, you can see information about software baked into the operating system and software you have installed. Other details are visible in this section as well, such as the **Environment Variables** and **Network Connections**.

Recall from the **Windows Fundamentals 1 room** (The Windows\System32 Folder task) where **Environment Variables** was briefly touched on.

Per Microsoft, "Environment variables store information about the operating system environment. This information includes details such as the operating system path, the number of processors used by the operating system, and the location of temporary folders.

The environment variables store data that is used by the operating system and other programs. For example, the WINDIR environment variable contains the location of the Windows installation directory. Programs can query the value of this variable to determine where Windows operating system files are located".

Click on **Environment Variables** to see the assigned values for the virtual machine.

Another method to view environment variables is **Control Panel > System and Security > System > Advanced system settings > Environment Variables** OR **Settings > System > About > system info > Advanced system settings > Environment Variables**.

The detour is over. Let's redirect our attention back to msinfo32 and pick up where we left off.

Towards the very bottom of this utility, there is a search bar. Please give it a go. Select Components and search for **IP address**.

Q1) What is the command to open System Information? (The answer is the name of the .exe file, not the full path)
Answer: msinfo32.exe

Q2) What is listed under System Name?
Answer: THM-WINFUN2

Q3) Under Environment Variables, what is the value for ComSpec?
Answer: %SystemRoot%\system32\cmd.exe

>> Task 6: Resource Monitor

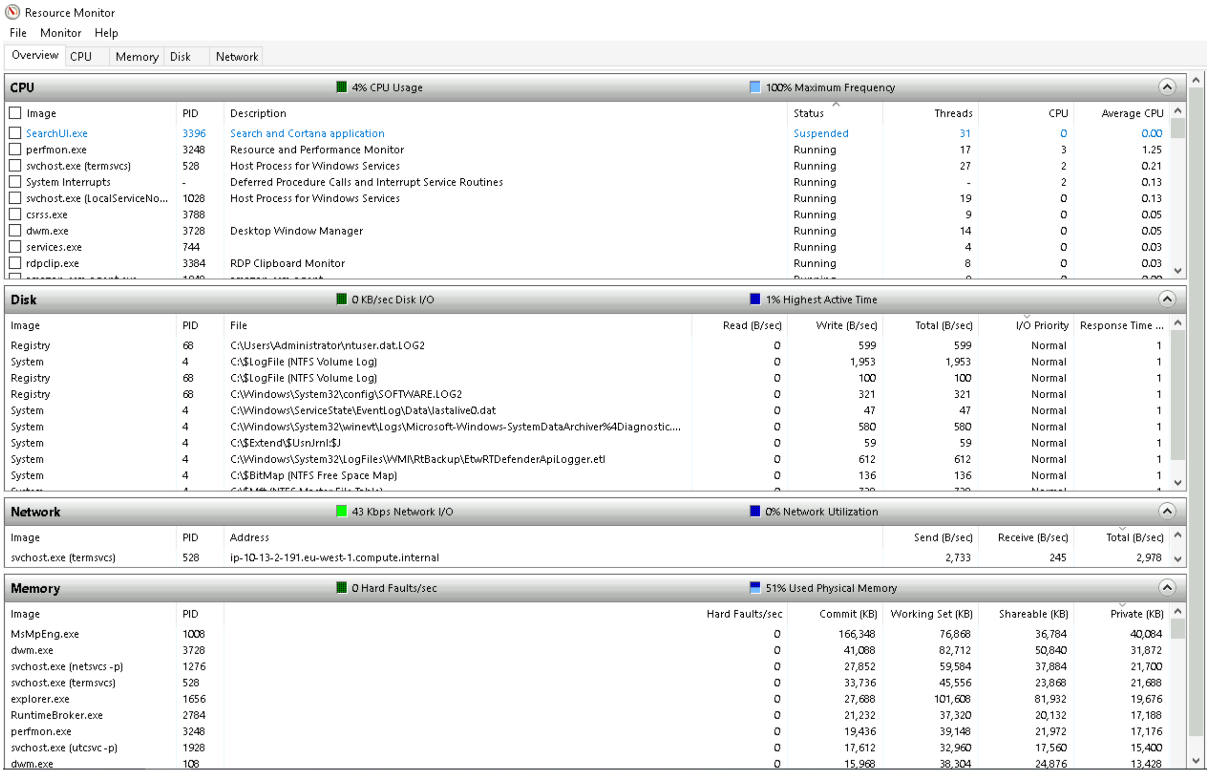
We're continuing with Tools that are available through the **System Configuration** panel.

What is **Resource Monitor** (resmon)?

Per Microsoft, "Resource Monitor displays per-process and aggregate CPU, memory, disk, and network usage information, in addition to providing details about which processes are using individual file handles and modules. Advanced filtering allows users to isolate the data related to one or more processes (either applications or services), start, stop, pause, and resume services, and close unresponsive applications from the user interface. It also includes a process analysis feature that can help identify deadlocked processes and file locking conflicts so that the user can attempt to resolve the conflict instead of closing an application and potentially losing data."

As some of the other tools mentioned in this room, this utility is geared primarily to advanced users who need to perform advanced troubleshooting on the computer system.

In the Overview tab, Resmon has four sections:
- **CPU**
- **Disk**
- **Network**
- **Memory**



The same four sections have corresponding tabs across the top.

Note that each tab has additional information for each.

Q1) What is the command to open Resource Monitor? (The answer is the name of the .exe file, not the full path)
Answer: resmon.exe

>> Task 7: Command Prompt

We're continuing with Tools that are available through the **System Configuration** panel.

The command prompt (cmd) can seem daunting at first, but it's really not that bad once you understand how to interact with it.

In early operating systems, the command line was the sole way to interact with the operating system.

When the GUI (graphical user interface) was introduced, it allowed users to perform complex tasks with a few clicks of a button instead of entering commands in the command prompt.

Even though the GUI is the primary way to interact with the operating system, a computer user can still interact via the command prompt.

In this task, we'll only cover a few commands that a computer user can run in the command prompt to obtain information about the computer system.

Let's start with a few simple commands, such as hostname and whoami.

The command hostname will output the computer name.

The command whoami will output the name of the logged-in user.

Next, let's look at some commands that are useful when troubleshooting.

A command used often is ipconfig. This command will show the network address settings for the computer.

Each command will have a help manual to explain the expected syntax to execute the command properly, along with any additional parameters that can be added to the command to expand its execution.

A command to retrieve the help manual for a command is /?.

For example, to see the help manual for ipconfig, you can use the following command: ipconfig /?

Note: To clear the command prompt screen, the command is cls.

The next command is netstat. Per the help manual, this command will display protocol statistics and current TCP/IP network connections.

netstat
netstat /?

The structure tells us the netstat command can be run alone or with parameters, such as -a, -b, -e, etc.

When any of the parameters are appended to the root command, netstat in this case, the output changes. Play with a few to see for yourself.

The net command is primarily used to manage network resources. This command supports sub-commands.

If you type net without a sub-command, the output will show the syntax for the root command showing a few of the sub-commands you can use.

```
C:\Users\Administrator>net
The syntax of this command is:

NET
    [ ACCOUNTS | COMPUTER | CONFIG | CONTINUE | FILE | GROUP | HELP |
      HELPMSG | LOCALGROUP | PAUSE | SESSION | SHARE | START |
      STATISTICS | STOP | TIME | USE | USER | VIEW ]
```

For the net command, to display the help manual /? will not work. In this case, you need to use different syntax, which is net help.

So, if you wish to see the help information for net user, the command is net help user.

You can use the same command to view the help information for other useful net sub-commands, such as localgroup, use, share, and session.

Refer to the following link to see a comprehensive list of commands you can execute in the command prompt here (https://ss64.com/nt/).

Q1) In System Configuration, what is the full command for Internet Protocol Configuration?
Answer: C:\Windows\System32\cmd.exe /k %windir%\system32\ipconfig.exe

Q2) For the ipconfig command, how do you show detailed information?
Answer: ipconfig /all

>> Task 8: Registry Editor

We're continuing with Tools that are available through the **System Configuration** panel.

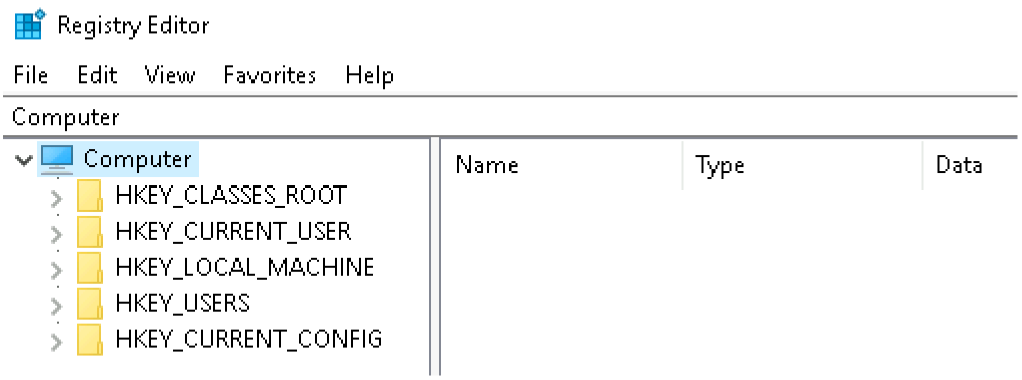
The **Windows Registry** (per Microsoft) is a central hierarchical database used to store information necessary to configure the system for one or more users, applications, and hardware devices.

The registry contains information that Windows continually references during operation, such as:

- Profiles for each user
- Applications installed on the computer and the types of documents that each can create
- Property sheet settings for folders and application icons
- What hardware exists on the system
- The ports that are being used.

Warning: The registry is for advanced computer users. Making changes to the registry can affect normal computer operations.

There are various ways to view/edit the registry. One way is to use the **Registry Editor** (regedit).



Refer to the following Microsoft documentation here (https://learn.microsoft.com/en-us/troubleshoot/windows-server/performance/windows-registry-advanced-users) to learn more about the Windows Registry.

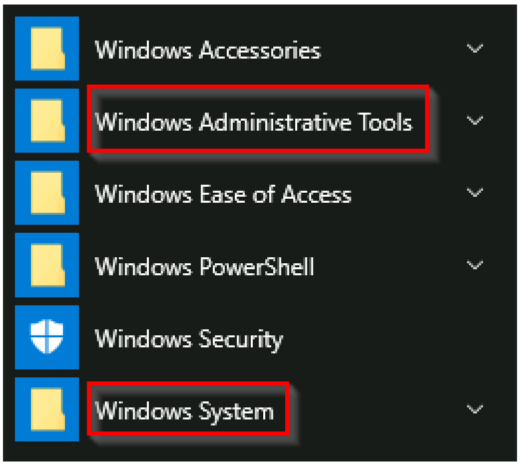
Q1) What is the command to open the Registry Editor? (The answer is the name of the .exe file, not the full path)
Answer: regedt32.exe

>> Task 9: Conclusion

Recall that the tasks covered in this room were some of the tools that can launch from **MSConfig**.

Throughout the room, commands and shortcuts were shared for the utilities. This means you don't have to launch **MSConfig** to run these utilities.

You can also run some of these utilities directly from the Start Menu.



Some of the tools listed in **MSConfig** that weren't mentioned in this room were either covered in Windows Fundamentals 1 or were left for you to explore on your own.